Monte Carlo Pricing of a European Call: Antithetic Variates

Quant Finance Portfolio — Phase 2, Block 2, Algorithm 1

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Abstract

We apply the antithetic variates technique of the Block 2.0 writeup to the Monte Carlo pricer of the European call under the exact GBM sampler of Block 1.1. The monotonicity condition holds trivially, so the AV estimator is guaranteed to reduce variance over the IID estimator at equal sample size, and to reduce VxT over the IID estimator at equal computational budget. We predict the variance reduction factor analytically as a function of moneyness (S_0/K) and design the empirical validation that measures the factor and the VxT ratio.

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1 Introduction

Block 1.1 produced our baseline European-call pricer:

$$\hat{C}_M^{\mathrm{IID}} \;=\; \frac{1}{M}\sum_{i=1}^{M} e^{-rT}(S_T^{(i)}-K)^+, \quad S_T^{(i)} = S_0\exp\big((r-\tfrac{1}{2}\sigma^2)T+\sigma\sqrt{T}\,Z_i\big), \tag{1}$$

with $Z_i \sim \mathcal{N}(0,1)$ i.i.d. The empirical variance at canonical parameters ($S_0 = K = 100$, $r = 0.05$, $\sigma = 0.20$, $T = 1$) is $\sigma_Y^2 \approx 217$, giving a half-width of about 0.09 at $M = 10^5$.

In this block we replace this estimator with its antithetic-variates counterpart, derive the resulting variance under GBM, predict the reduction factor as a function of moneyness, and validate empirically.

2 The antithetic estimator for the European call

2.1 Construction

Let $f(z) \stackrel{\text{def}}{=} e^{-rT}(S(z) - K)^+$ where

$$S(z) \stackrel{\text{def}}{=} S_0 \exp\big((r - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\, z\big), \qquad z \in \mathbb{R}. \tag{2}$$

The antithetic-pair payoff is

$$Y_i^{\text{AV}} \stackrel{\text{def}}{=} \tfrac{1}{2}\big(f(Z_i) + f(-Z_i) \big), \tag{3}$$

with $Z_i \sim \mathcal{N}(0,1)$ i.i.d. The estimator is

$$\hat{C}_M^{\text{AV}} \stackrel{\text{def}}{=} \frac{1}{M} \sum_{i=1}^{M} Y_i^{\text{AV}}. \tag{4}$$

Each Y_i^{AV} requires *two* payoff evaluations (one at Z_i, one at $-Z_i$), so the per-pair computational cost is roughly twice that of an IID payoff. Per the VxT criterion of Block 2.0, the relevant comparison is between $\hat{C}_M^{\text{AV}}$ (using $2M$ payoff evaluations) and $\hat{C}_{2M}^{\text{IID}}$ (also using $2M$ payoff evaluations).

2.2 Verification of monotonicity

The function $z \mapsto S(z)$ is strictly increasing in z (it is S_0 times a strictly increasing exponential). The function $s \mapsto e^{-rT}(s - K)^+$ is monotone non-decreasing in s, strictly increasing on $\{s > K\}$. The composition $z \mapsto f(z)$ is therefore monotone non-decreasing, strictly increasing on $\{z : S(z) > K\}$.

By Theorem 3.2 of the Block 2.0 writeup, $\text{Cov}(f(Z), f(-Z)) \le 0$ with strict inequality (since f is not essentially constant: the ITM and OTM regions both have positive probability under the standard normal). The antithetic estimator is therefore strictly superior to the IID estimator at equal sample size.

2.3 Variance and reduction factor

By Theorem 3.1 of the Block 2.0 writeup,

$$\text{Var}(\hat{C}_M^{\text{AV}}) \;=\; \frac{1}{2M}\big(\sigma_f^2 + \text{Cov}(f(Z), f(-Z)) \big), \qquad \sigma_f^2 \stackrel{\text{def}}{=} \text{Var}(f(Z)). \tag{5}$$

At equal computational budget,

$$\text{Var}(\hat{C}_{2M}^{\text{IID}}) \;=\; \frac{\sigma_f^2}{2M}, \tag{6}$$

so the variance reduction factor relative to IID-at-equal-cost is

$$\text{VRF} \stackrel{\text{def}}{=} \frac{\text{Var}(\hat{C}_{2M}^{\text{IID}})}{\text{Var}(\hat{C}_M^{\text{AV}})} \;=\; \frac{\sigma_f^2}{\sigma_f^2 + \text{Cov}(f(Z), f(-Z))} \;=\; \frac{1}{1 + \text{Corr}(f(Z), f(-Z))}. \tag{7}$$

The VRF is exactly 1 if $f(Z)$ and $f(-Z)$ are uncorrelated (no benefit), goes to ∞ as the correlation approaches -1 (perfect negative correlation, infinite reduction), and equals $1/2$ if the correlation approaches $+1$ (AV is twice as bad as IID — but this cannot happen here, by the monotonicity result).

In particular, VRF > 1 if and only if $\text{Corr}(f(Z), f(-Z)) < 0$, which is exactly what monotonicity guarantees.

3 Predicting the VRF as a function of moneyness

How large is the correlation $\mathrm{Corr}(f(Z), f(-Z))$ for the European call under GBM? The answer depends on moneyness, and the dependence is informative.

3.1 Two limiting regimes

Deep ITM ($K \to 0$): the call payoff approaches the forward, $f(z) \to e^{-rT}S(z) - e^{-rT}K \cdot e^{rT} = e^{-rT}S(z) - Ke^{-rT}$. (For ATM and ITM cases below the strike K appears as a constant after discounting.) The dominant variation is in $S(z)$, which is monotone in z in a strict, smooth way. The function is essentially $z \mapsto a \cdot e^{bz}$ for constants a, b. We expect $f(Z)$ and $f(-Z)$ to have strong negative correlation (near -1), giving very high VRF.

Deep OTM ($K \to \infty$): the call payoff is zero on $\{S(z) < K\}$, which is the overwhelming majority of paths. Both $f(Z)$ and $f(-Z)$ are zero with high probability, and one is nonzero only when the other is zero (in opposite directions of z). The correlation is small in absolute terms (close to zero) because both samples are zero most of the time, giving VRF close to 1.

ATM ($K = S_0$): the regime of practical interest. The call activates roughly half the time, with the corresponding negative correlation under reflection. We expect a moderate negative correlation, around -0.5 to -0.7.

3.2 The kink and the loss of correlation

The reason the call payoff has *less than perfect* antithetic correlation, even though it is monotone, is the kink at $S = K$. On the ITM side (z large positive), $f(z) = e^{-rT}(S(z) - K)$ varies smoothly and approximately linearly in $S(z)$. On the OTM side (z large negative), $f(z) = 0$, identically. The reflection $z \mapsto -z$ maps the ITM side to the OTM side and vice versa, but the OTM side carries no information (zero), so the negative correlation cannot be perfect.

In contrast, an option whose payoff is smooth and strictly monotone everywhere (such as a plain forward, $f(z) = S(z) - F$) achieves $\mathrm{Corr}(f(Z), f(-Z))$ very close to -1. The kink in the call payoff caps the achievable correlation strictly above -1.

For our running ATM example we predict VRF in the range VRF $\approx$ 1.5–3.0, which corresponds to negative correlations in the range -0.7 to -0.3. The exact number will emerge from the validation experiment of Section 5.

4 Algorithmic specification

Algorithm 1 Antithetic-variates Monte Carlo for the European call (exact GBM)

Require: Spot S_0, strike K, maturity T, rate r, volatility σ, sample size M, confidence level $1-\beta$, random generator `rng`.

Ensure: Estimate $\hat{C}^{\mathrm{AV}}$, half-width h_M^{AV}, sample variance $S_M^{2,\mathrm{AV}}$.

$D \leftarrow e^{-rT}, \quad \mu \leftarrow (r - \frac{1}{2}\sigma^2)T, \quad \nu \leftarrow \sigma\sqrt{T}$
for $i = 1, \ldots, M$ **do**
  Draw $Z_i \sim \mathcal{N}(0,1)$ using `rng`
  $S_+^{(i)} \leftarrow S_0 \exp(\mu + \nu Z_i)$
  $S_-^{(i)} \leftarrow S_0 \exp(\mu - \nu Z_i)$
  $Y_i^{\mathrm{AV}} \leftarrow \frac{1}{2} D \left(\max(S_+^{(i)} - K, 0) + \max(S_-^{(i)} - K, 0)\right)$
end for
$\hat{C}^{\mathrm{AV}} \leftarrow M^{-1} \sum_i Y_i^{\mathrm{AV}}$
$S_M^{2,\mathrm{AV}} \leftarrow (M-1)^{-1} \sum_i (Y_i^{\mathrm{AV}} - \hat{C}^{\mathrm{AV}})^2$
$h_M^{\mathrm{AV}} \leftarrow z_{\beta/2} \sqrt{S_M^{2,\mathrm{AV}}/M}$
return $(\hat{C}^{\mathrm{AV}}, h_M^{\mathrm{AV}}, S_M^{2,\mathrm{AV}})$

The algorithm uses the same standard normal Z_i to drive both elements of the antithetic pair, exploiting that $S(\nu Z) = S_0 e^{\mu+\nu Z}$ and $S(-\nu Z) = S_0 e^{\mu-\nu Z}$ require only the sign flip in the exponent. No additional random draws are needed beyond the M standard normals already used by the IID estimator.

The variance and half-width are computed from the *paired payoffs* Y_i^{AV}, not from the individual $f(Z_i)$ and $f(-Z_i)$. This is essential: the Y_i^{AV} are i.i.d. in i (each pair is independent of the rest), and `mc_estimator` applied to them returns the correct variance and half-width automatically. Treating the $2M$ individual payoff evaluations as i.i.d. would be wrong: $f(Z_i)$ and $f(-Z_i)$ are highly correlated, and the independence assumption would inflate the apparent half-width.

This is structurally important: *the variance-reducing correlation lives inside each pair, not across the dataset*, which is what makes the AV estimator both correct and efficient.

5 Empirical validation strategy

The validation script for the AV pricer comprises four tests.

Test 1: variance reduction factor. Estimate the empirical correlation $\hat{\rho} \stackrel{\text{def}}{=} \widehat{\mathrm{Corr}}(f(Z), f(-Z))$ from a large sample of M standard normals, and compute the VRF

$$\widehat{\mathrm{VRF}} = \frac{1}{1+\hat{\rho}}. \tag{8}$$

For our canonical ATM call, the prediction is $\hat{\rho} \in [-0.7, -0.3]$, $\widehat{\mathrm{VRF}} \in [1.4, 3.3]$. The test passes if the empirical correlation is negative and the VRF is meaningfully above 1.

Test 2: consistency with the IID estimator. The AV estimator and the IID estimator should agree on the same path data (when computed from the same RNG state, which they will in our design). Concretely: at $M = 10^4$ pairs, run AV pricer and IID pricer with $M_{\mathrm{IID}} = 2M$ paths from the same seed, and verify that both estimates fall within a few half-widths of the closed-form Black-Scholes price. (We do not require pathwise equality of the two estimates

because the path constructions differ: the AV pricer uses each Z_i for both elements of the pair, while the IID pricer would treat them as $2M$ independent draws.)

Test 3: VxT comparison at fixed budget. Run both estimators with the same number of payoff evaluations ($2M$ each) and report:

- Empirical sample variance of Y^{AV} vs sample variance of Y^{IID} at the same accumulated number of payoffs.
- Half-widths of the two estimators.
- Ratio (which should equal $\widehat{\mathrm{VRF}}$ from Test 1 up to noise).

Wall-clock measurements are not necessary for the test to pass; the per-pair compute is dominated by two payoff evaluations and one shared random draw, with negligible overhead, so the VxT ratio equals the variance ratio to within rounding.

Test 4: input validation. Mechanical: the AV pricer rejects bad inputs and accepts negative interest rates.

6 Bibliographic notes

The textbook reference is Glasserman [2, Section 4.2]. The general theorem on antithetic variates and monotone functions goes back to Hammersley & Morton; the application to financial Monte Carlo is in Boyle, Broadie & Glasserman [1].

References

[1] P. Boyle, M. Broadie, and P. Glasserman, *Monte Carlo methods for security pricing*, Journal of Economic Dynamics and Control, 21(8–9):1267–1321, 1997.

[2] P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2004.